25900362 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEECT204 - SMART SENSORS WITH IOT

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Which type of sensor is used for burglar alarms? [U] / [CO1]

   a). Ultrasonic sensor b). Gas sensor

   c). PIR sensor d). Voltage sensor

2. Which application requires accurate current sensing? [U] / [CO1]

   a). Lighting a bulb b). Smart energy meters

   c). Measuring sound d). Recording video

3. Which of the following is NOT a type of potentiometer? [K] / [CO1]

   a). Rotary b). Capacitive

   c). Digital d). Linear

4. Which application uses both IR and piezoelectric sensors? [U] / [CO1]

   a). Smartphone touch systems b). Weather monitoring

   c). Power storage d). Magnetic data storage

5. Piezoelectric sensors generate ____ [U] / [CO1]

a). Voltage from stress b). Light from vibration

c). Heat from sound d). Magnetism from current

6. What happens to the output of an R-2R DAC if all digits inputs are 0? [U] / [CO2]

a). maximum voltage b). half the refernece voltage c). zero volts d). unpredictable

7. Which of the following is a disadvantage of semiconductor strain gauge? [U] / [CO2]

a). High temperature sensitivity b). Low temperature sensitivity

c). Linear behavior d). High Fatigue Life

8. The full form of LVDT is [U] / [CO2]

a). Linear Variable Displacement Transducer b). Linear Value Displacement Transformer

c). Linear Variable Display Transformer d). Linear Variable Displacement Type

9. In SAR ADC, what happens if the comparator makes an incorrect decision at any step? [U] / [CO2]

a). The system resets b). Only the affected bit is wrong

c). The entire digital output is incorrect d). Conversion continues unaffected.

10. What is the primary purpose of interfacing a transducer with a microcontroller or data acquisition system? [U] / [CO2]

a). To increase voltage b). To measure temperature directly

c). To convert sensor signals into a compatiable format d). To reduce resistance

11. Solenoids are commonly used in [An] / [CO3]

a). Electric bells b). Door locks c). Valves d). All of the above

12. A stepper motor is classified as which type of actuator [An] / [CO3]

a). Linear actuator | b). Rotary actuator | c). Hydraulic actuator | d). Pneumatic actuator

13. In a DC motor interfacing circuit, PWM duty cycle of 75% means: [An] / [CO3]

a). Motor rotates at half speed

b). Motor rotates at three-fourth of maximum speed

c). Motor rotates at full speed

d). Motor rotates in reverse

14. Which component is commonly used to drive a DC motor from a microcontroller? [An] / [CO3]

a). Resistor

b). H-Bridge driver

c). Capacitor

d). Inductor

15. Which of the following is a type of servomotor? [An] / [CO3]

a). AC servomotor | b). DC servomotor | c). Stepper motor | d). All of the above

16. Which of the following is NOT a layer in industrial automation architecture? [U] / [CO4]

a). Field Level

b). Control Level

c). Enterprise Level

d). Application Programming Level

17. The main source of power in Industry 2.0 was: [U] / [CO4]

a). Steam

b). Electricity

c). Wind

d). Solar

18. Which key is required for uploading (writing) light intensity data to ThingSpeak? [U] / [CO4]

a). Read API key

b). Write API key

c). Channel ID

d). Encryption key

19. An IoT-enabled temperature sensor in smart homes is an example of: [U] / [CO4]

a). Traditional sensor

b). Smart sensor

c). Analog device

d). Passive component

20. Which function is commonly used in Arduino to send HTTP requests to ThingSpeak? [U] / [CO4]

a). Serial.begin() b). ThingSpeak.writeField()

c). delay() d). analogRead()

[1x20 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

21. Why is filtering important in signal conditioning? [U] / [CO1]

22. List industrial applications of pressure sensors. [K] / [CO1]

23. What are the main criteria for selecting transducers? [K] / [CO2]

24. Discuss one advantage and one disadvantage of active transducers. [U] / [CO2]

25. How do linear actuators contribute to automation in the food and beverage industry? Give examples. [An] / [CO3]

26. Give three examples of relay applications in automobiles. [An] / [CO3]

27. Why is a relay module required for interfacing a water pump with ESP32? [U] / [CO4]

28. Explain how industrial automation contributes to worker safety. [U] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

29. Discuss the need for sensors and transducers in modern industry with suitable examples. [U] / [CO1]

30. Explain the advantages and limitations of LM358. Also list the applications of LM358. [U] / [CO2]

31. Discuss the different types of linear actuators with examples. [An] / [CO3]

32. Design a block diagram of an IoT-enabled automatic fan control system. Explain each component and its function in detail. [U] / [CO4]

[10x2 = 20]